

The Chain of infection

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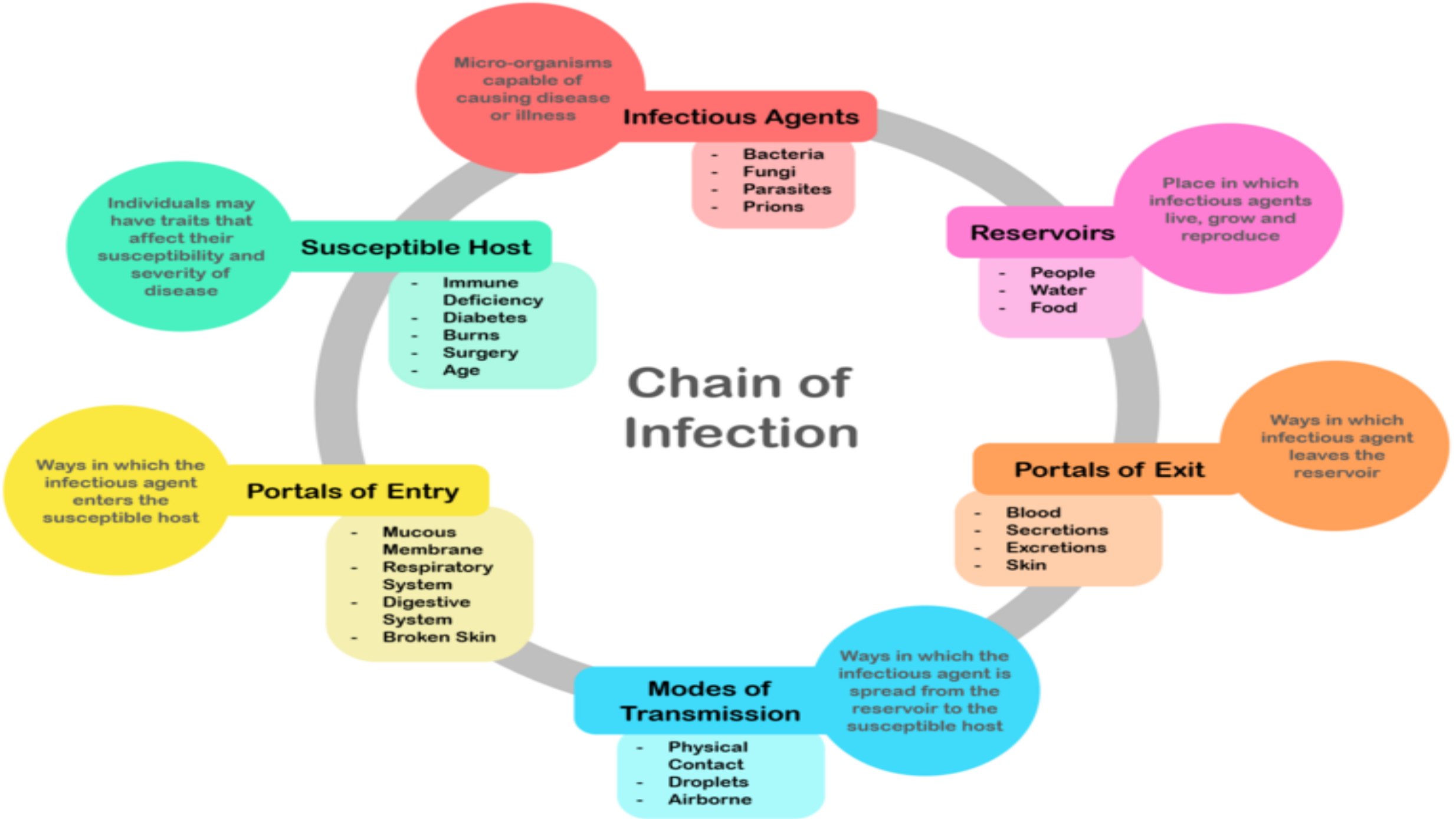
Infection control consultant

NLI

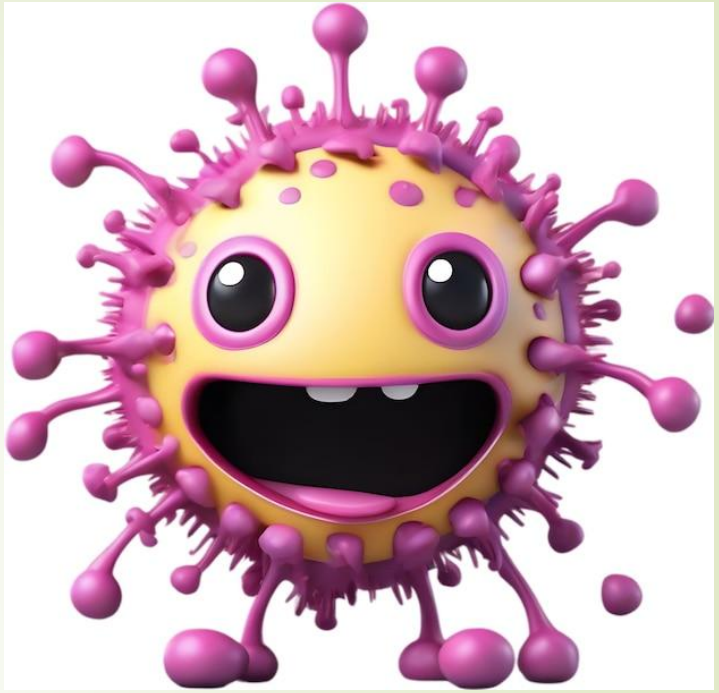
- **the process by which a pathogen spreads from one host to the next**

- There are six steps in the chain of infection .and transmission will only take place if all six links are intact.




Chain of Infection
Micro-organisms capable of causing disease or illness
Infectious Agents
- Bacteria
- Fungi
- Parasites
- Prions
Place in which infectious agents live, grow and reproduce
Reservoirs
- People
- Water
- Food
Ways in which infectious agent leaves the reservoir
Portals of Exit
- Blood
- Secretions
- Excretions
- Skin
Ways in which the infectious agent is spread from the reservoir to the susceptible host
Modes of Transmission
- Physical Contact
- Droplets
- Airborne
Ways in which the infectious agent enters the susceptible host
Portals of Entry
- Mucous Membrane
- Respiratory System
- Digestive System
- Broken Skin
Individuals may have traits that affect their susceptibility and severity of disease
Susceptible Host
- Immune Deficiency
- Diabetes
- Burns
- Surgery
- Age

The pathogen (infectious agent)



- The first link in the chain of infection
- Viruses
- Bacteria
- Fungi
- Parasitic
- Prions
- How well any pathogen is able to thrive depends on three factors
  - Its **pathogenicity –** its ability to produce disease
  - Its degree of **virulence** – its severity or harmfulness
- Its **invasiveness** – its tendency to spread

The reservoir



- A reservoir is the principal habitat in which a pathogen lives, flourishes and is able to multiply.
- Human reservoirs
- Animal & insect reservoirs
- Environmental reservoirs

Human reservoirs

- **acute clinical cases:**

someone is infected and is displaying signs and symptoms of the disease

- **carriers**

someone has been colonized with an infectious agent but is not unwell

Which one is more dangerous?

The carrier



Types of carrier:

- **Incubatory carriers** – people who are infectious even before their own symptoms start ex:chicken pox
- **Inapparent (healthy)carriers** – in which an individual is able to transmit an infection to others, without ever developing the infection themselves ex:typhoid
- **Convalescent carriers** – people who are in the recovery phase of their illness but who continue to be infectious for a short-term days or weeks ex:measles
- **Chronic carriers** – anyone who has recovered but who continues to be a carrier for infection for long-term, years and even the entire life. Ex:HBV&HCV



Animal & insect reservoirs

- Any infectious disease that is transmitted under natural conditions from animal to human is referred to as **zoonosis.**
- Rabies (which is transmitted by dogs, cats, foxes and bats bites)
- Lyme disease (which is transmitted via ticks)
- Early Covid 19(which is transmitted by eating bats)



Environmental reservoirs

- soil (which acts as a reservoir for Clostridium tetani, the causative agent of tetanus)
- Water,air-conditioning systems, swimming pools . (reservoir for Legionella pneumophila, the causative agent of Legionnaire's disease).
- Moist places such as sink-traps, drains and flower vases , contact lens soloutions & even antiseptic solutions can harbour *Pseudomonas.*

The portal of exit



- The portal of exit is any route which enables a pathogen to leave the reservoir or host.
  - **Alimentary** – via vomiting, diarrhoea or biting
  - **Genitourinary** – via sexual transmission
  - **Respiratory** – through coughing, sneezing and talking
  - **Skin** – via skin lesions
  - **Trans-placental** – where transmission is from mother to foetus

The mode of transmission



- **A) Direct Transmission**:

occurs when the disease agent contacts a vulnerable host directly, without the need of an intermediary host, objects, or substance.

  - **Direct Contact with mm or skin**(STDs, HIV/AIDS, scabies, skin and eye infections)
  - **Contact with Contaminated Soil**(tetanus)
  - **Droplet Infection(**during coughing, sneezing, speaking or spitting.as common cold)
  - **Inoculation into Skin or Mucosa**

    rabies virus by dog bites, hepatitis B through contaminated needles and syringes
  - **Transplacental (**vertical)(TORCH)

b)Indirect Transmission

the 5Fs Plus Air Borne Transmission

the 5Fs

- Flies (vecror born) ex:plague
- Fingers (typhoid fever, diarrhea, streptococcal infectio
- Fomites (eye and skin infections)
- Food (salmonella ,shigella)
- Fluids (cholera)



Air Borne Transmission

contaminated infected nuclei of dust will be floating in the air

Ex: SARS, measles, tuberculosis

The portal of entry

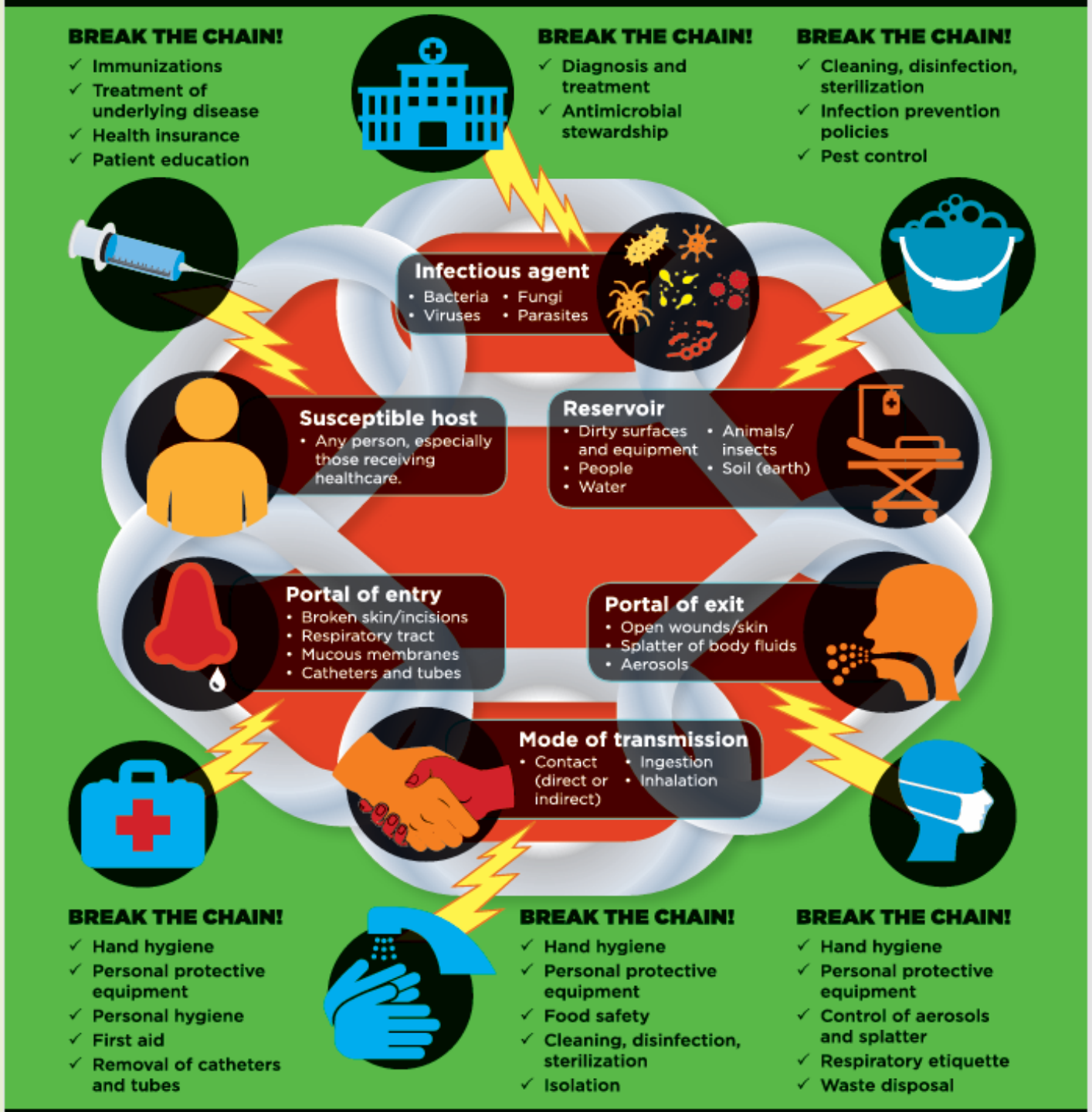


- The portal of entry is the means by which an infection is able to enter a susceptible host.
  - **Inhalation (**via the respiratory tract)
  - **Absorption** (via mucous membranes such as the eyes)
  - I**ngestion** (via the gastrointestinal tract)
  - **Inoculation** (as the result of an inoculation injury)

The susceptible host

- How susceptible any host will be, depends on a variety of factors:
  - Their **age** – and in particular if they are very young or very old
  - presence of **malnutrition or dehydration**
  - Whether there is any underlying chronic **disease**
  - If the host suffers from **immobility**
  - If they are taking any **medication** which could disrupt or suppress their immune response
  - **General resistance factors** (such as mucous membranes, skin, cough reflex etc) that can help defend against infection


BREAKING
THE CHAIN OF
INFECTION


BREAK THE CHAIN!
✓ Immunizations
✓ Treatment of underlying disease
✓ Health insurance
✓ Patient education
BREAK THE CHAIN!
✓ Diagnosis and treatment
✓ Antimicrobial stewardship
BREAK THE CHAIN!
✓ Cleaning, disinfection, sterilization
✓ Infection prevention policies
✓ Pest control
Infectious agent
• Bacteria • Fungi
• Viruses • Parasites
Susceptible host
• Any person, especially those receiving healthcare.
Reservoir
• Dirty surfaces and equipment • Animals/ insects
• People • Soil (earth)
• Water
Portal of entry
• Broken skin/incisions
• Respiratory tract
• Mucous membranes
• Catheters and tubes
Portal of exit
• Open wounds/skin
• Splatter of body fluids
• Aerosols
Mode of transmission
• Contact (direct or indirect) • Ingestion
• Inhalation
BREAK THE CHAIN!
✓ Hand hygiene
✓ Personal protective equipment
✓ Personal hygiene
✓ First aid
✓ Removal of catheters and tubes
BREAK THE CHAIN!
✓ Hand hygiene
✓ Personal protective equipment
✓ Food safety
✓ Cleaning, disinfection, sterilization
✓ Isolation
BREAK THE CHAIN!
✓ Hand hygiene
✓ Personal protective equipment
✓ Control of aerosols and splatter
✓ Respiratory etiquette
✓ Waste disposal

Health care-associated infections (HCAIs)

- are infections that occur while receiving health care, developed in a hospital or other health care facility that **first appear 48 hours or more** after hospital admission, **or within 30 days** after having received health care.



Hospital acquired infections are associated with

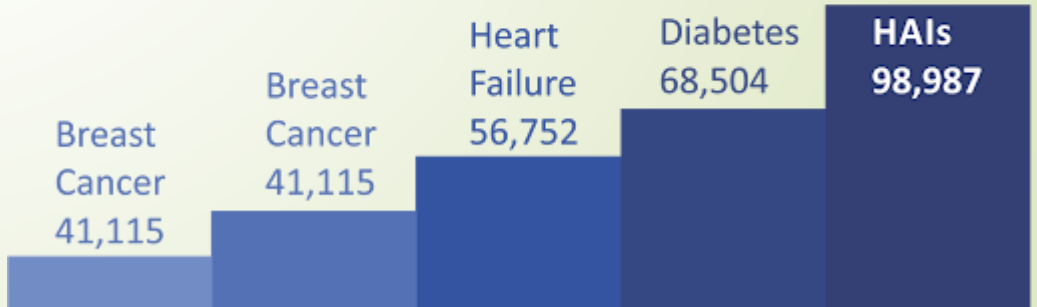
$96–$147 billion
direct and indirect costs in U.S. acute-care hospitals annually[2]

$10,375
additional in-hospital care costs per patient[3]

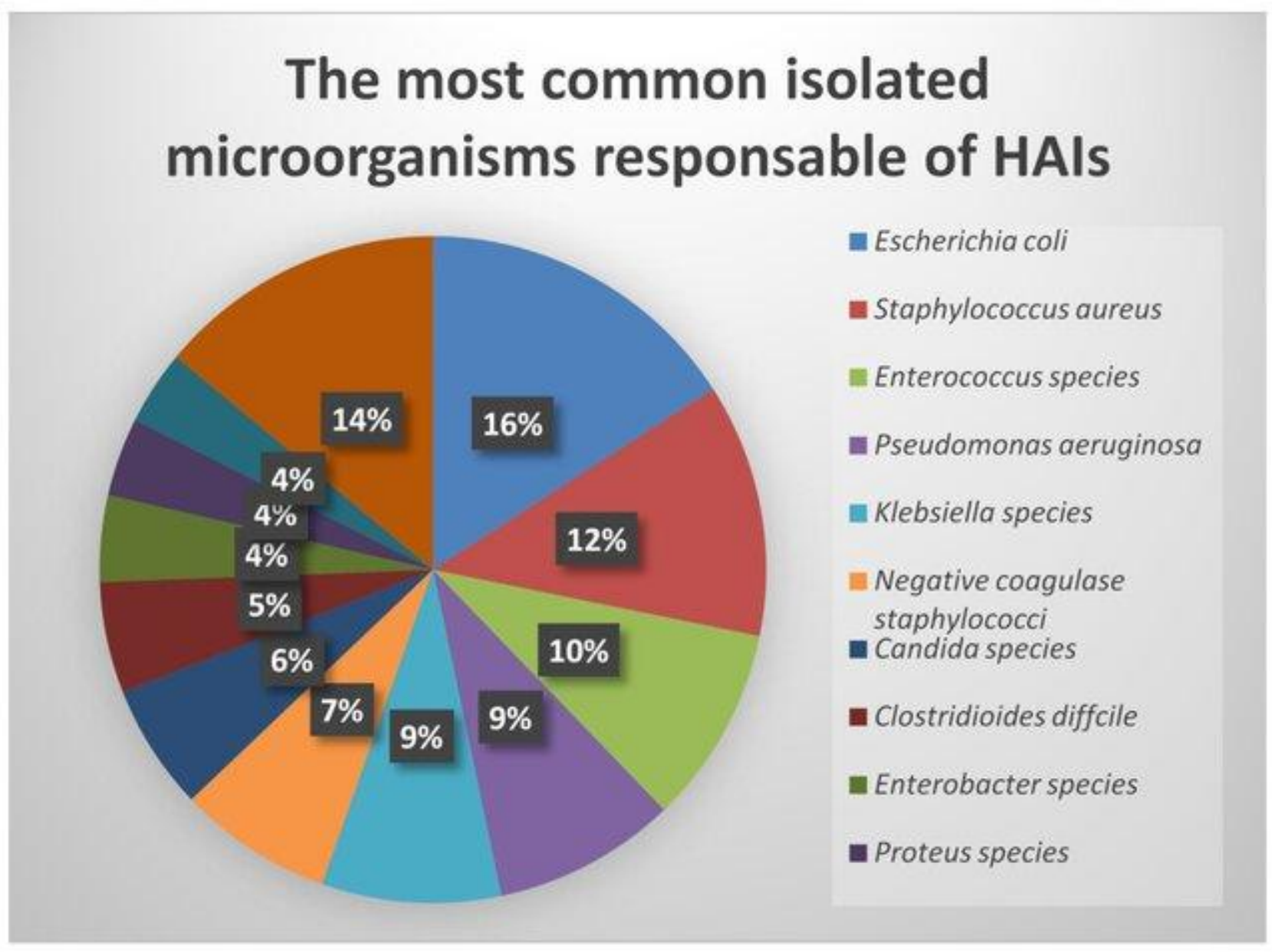
3.30 days
added to the length of stay[3]

1.7 million people per year an infection during a hospital stay

HAIs kill more people each year in USA than Breast Cancer and Prostate Cancer combined.




80
95
40
PHILIPS


The most common isolated microorganisms responsable of HAIs
16%
12%
10%
9%
9%
7%
6%
5%
4%
4%
4%
14%
Escherichia coli
Staphylococcus aureus
Enterococcus species
Pseudomonas aeruginosa
Klebsiella species
Negative coagulase staphylococci
Candida species
Clostridioides diffcile
Enterobacter species
Proteus species